Assignment Week 2

Footprinting and Network scanning

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Modules	Kali tools for footprinting and Zenmap
Client	Networkwalks
Permission	yes

Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools and scanning my own local network with Zenmap . First module covers the footprinting phase on networkwalks so we can gather public information to mapping live hosts on a network. All commands were run in Kali Linux (footprinting). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

Tools Used

The table below lists each tool used in this report and its purpose.

Tools	Use
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details.
whatweb	Fingerprint web technologies .
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
Zenmap	Scan the local subnet to find live hosts, IPs and MAC addresses.

Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using five Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl and Wafw00f . Each tool was used to collect a different type of information about the target.

WHOIS is for to obtain publicly available domain registration information and identify the domain’s name servers. It’s expiration date and update date some more information.**WhatWeb** to identify tools and software used for the website. The results identified WordPress 7.1.4 and Jquery 3.7.1, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified 192.232.216.135.
Server: 192.168.100.1
Address: 192.168.100.1#53

Curl is used with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.
Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs**)**.

Network Scanning with Zenmap

I used **Zenmap** is time to perform network discovery on my local network. For this first I have to know my ip, so I type ipconfig on the terminal to identify my local ip and subnet. I then entered the subnet into Zenmap and selected Ping Scan to identify active hosts.

The example results provided in the practical identified four live hosts:

- 10.0.0.1
- 10.0.0.2
- 10.0.0.3

The example results also included four MAC addresses.

After completing the scan, I opened the Topology section in Zenmap. After that to save Topology, I click the save Graphic option on the screen, name the file, save as pdf.

Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and Jquery	Attackers may use exposed version information to identify software requiring further security review	**Low**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location	**Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web firewall	**Low**
6	Multiple live hosts visible on local network	Zenmap identified three live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	**Medium**

Suggestion

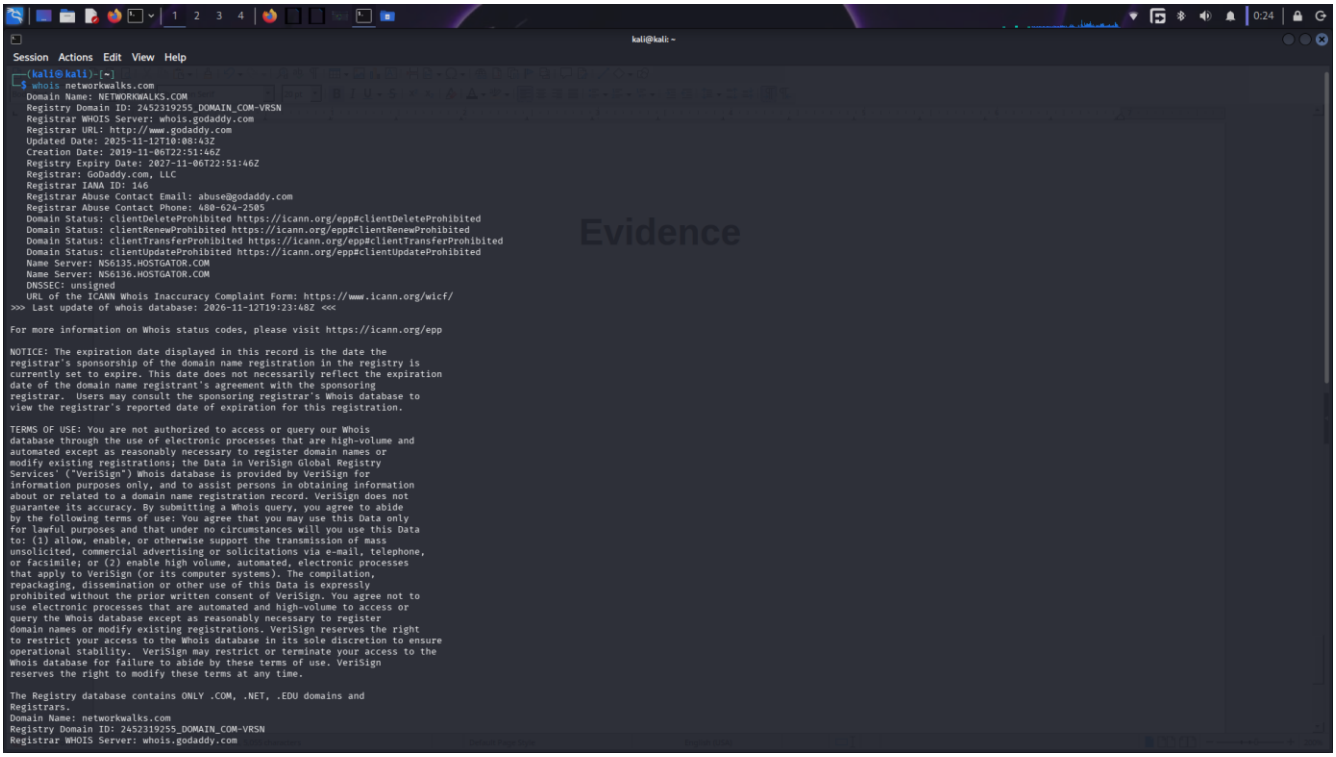
Based on the observations from these activities, I recommend the following security improvements:

1. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories. Now a days, Jquery have update available so I suggest to update Jquery . Now wordpress is okay due to in it's latest version
2. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
3. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
4. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
5. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.

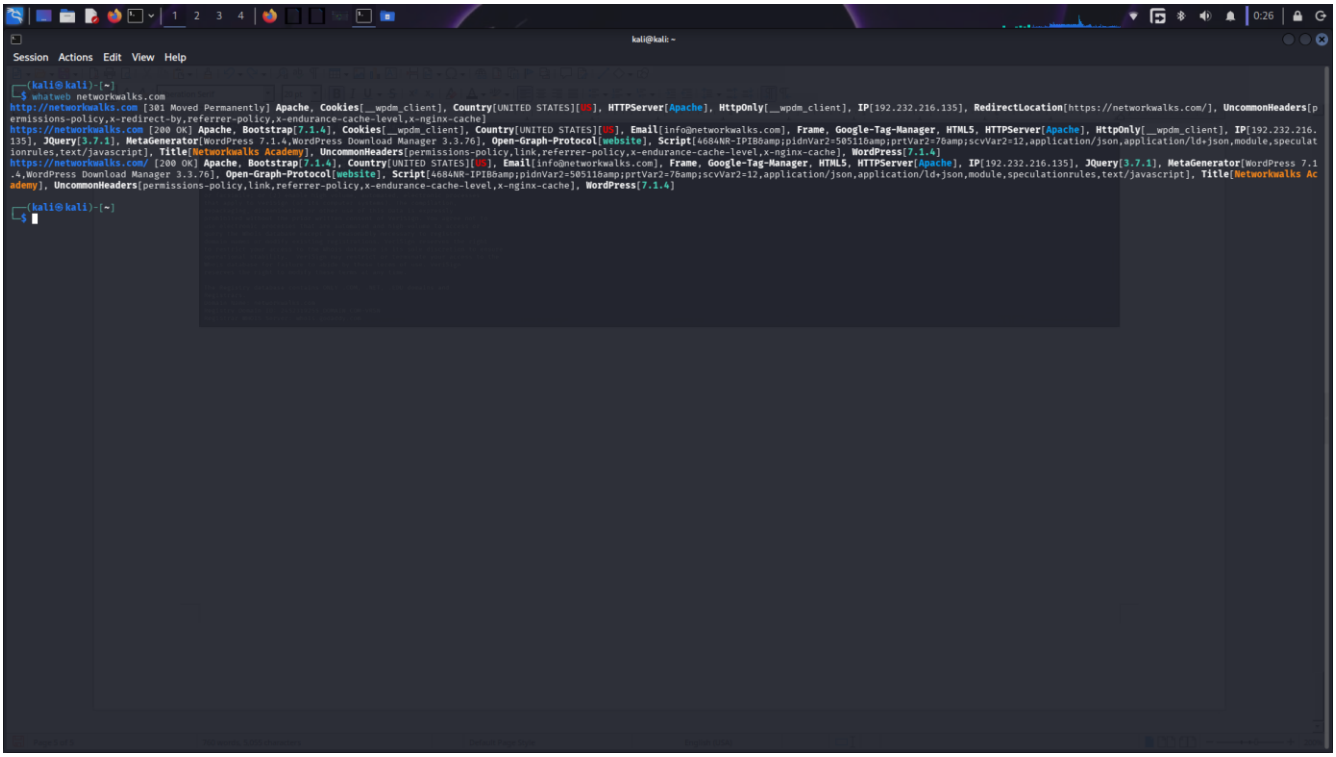
6. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
7. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
8. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

Evidence

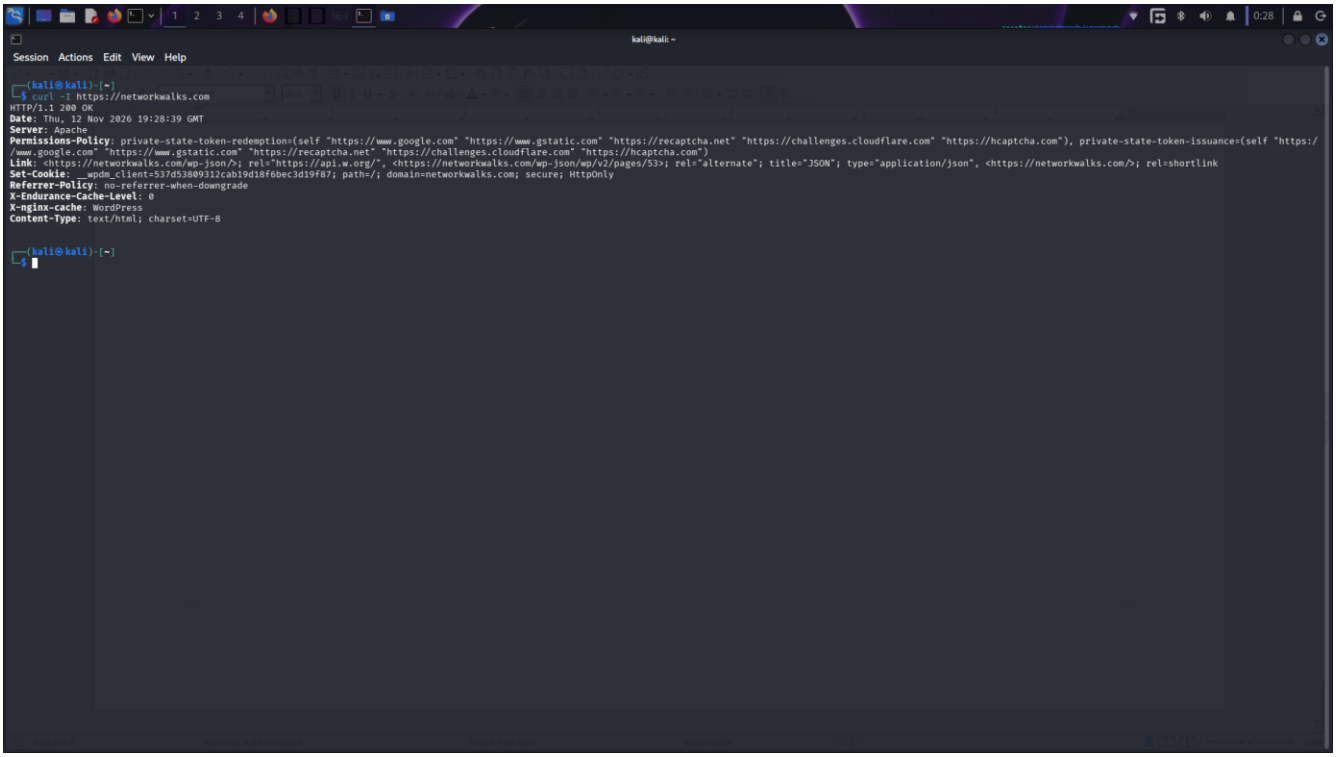
Whois



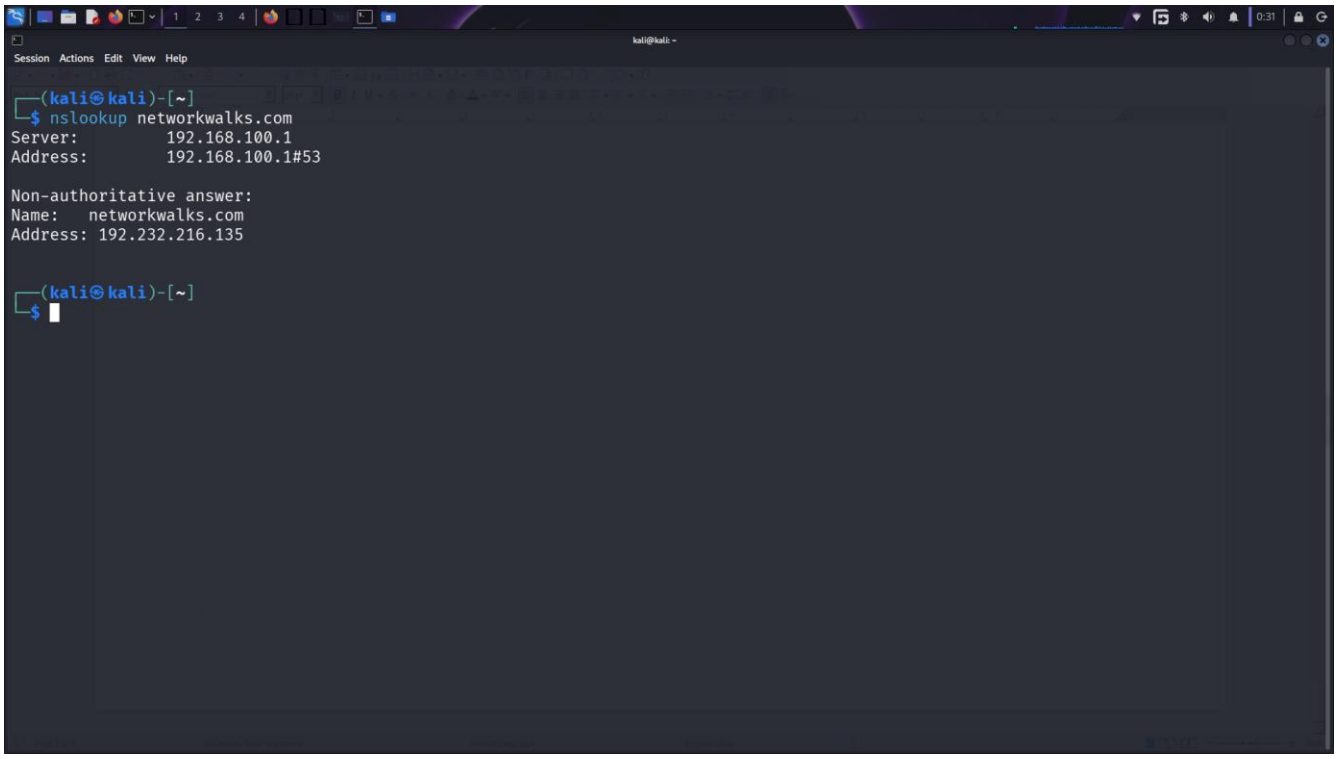
Whatweb



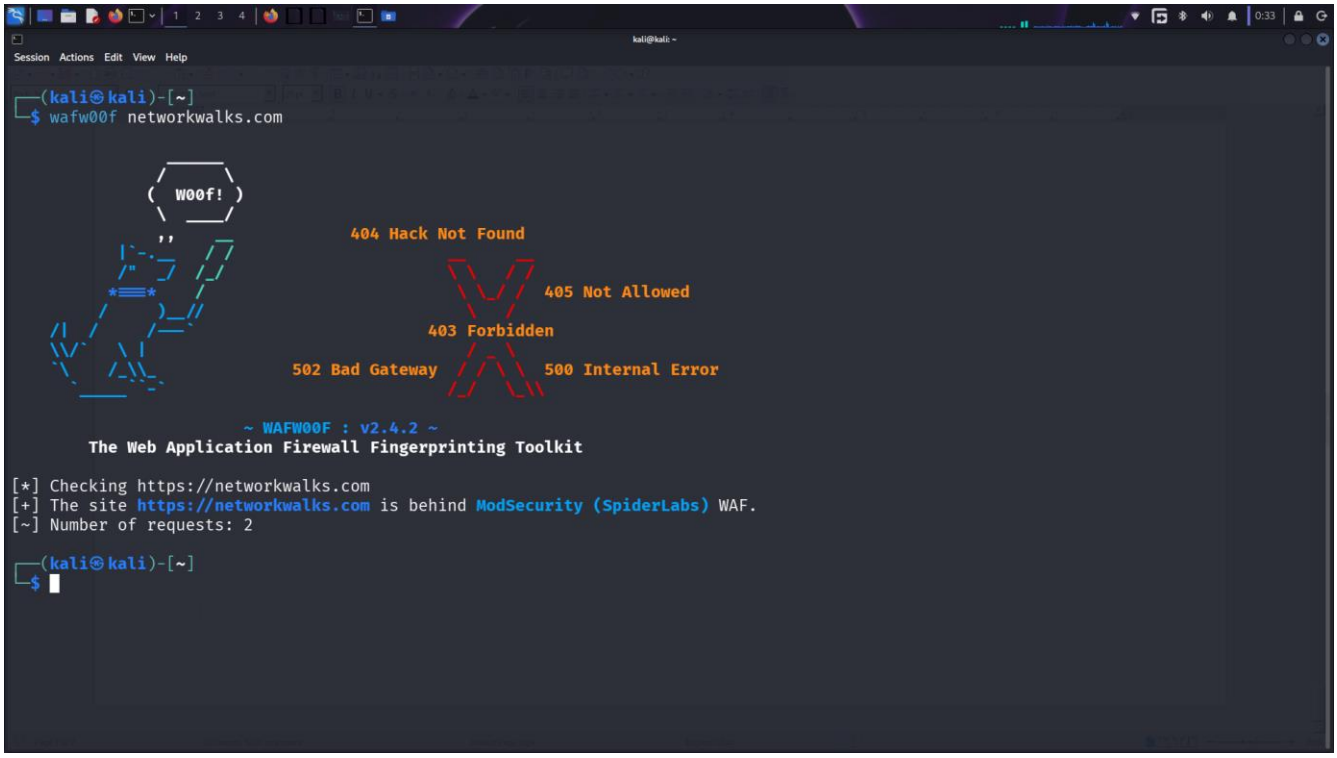
curl



nslookup



wafw00f



Local Network Scan

